RIN RAY TEACHING STATEMENT

rin.io · they/them

Years ago I gave two strangers a ride home after Thanksgiving, a ten-year-old girl and her father. From the back seat she told me she was not good at math. She didn't understand why equations work, and she was bad at multiplication. A few questions later it turned out she loved stories and patterns and spent her free time imagining shapes moving around. What she had been taught as mathematics was the spelling of it, and nobody had shown her the literature. I think about her whenever I teach.

I came into academia from outside, through an early-college program and a degree in computational physics at sixteen, and I go out of my way to present mathematics so that anyone can get in. For me mathematics is visual, interactive, intuitive and artistic, and above all it is a story. Three principles shape how I teach.

DISCOVERY BEFORE DEFINITION

I introduce material as though we, the class, are discovering it together: examples and history first, definitions and theorems afterwards, the order in which the mathematics was actually found. Euler plays with examples for pages before stating the theorem those patterns forced on him, and I want students to have that experience rather than receive its end product.

In linear algebra, which I have taught several times (as a teaching assistant for Northwestern's honors sequence in Intensive Linear Algebra and Multivariable Calculus, and for Linear Algebra), I start with images. Students use an interactive explanation, The Magnificent 2D Matrix, to see how changing a matrix stretches, moves and rotates a picture, which needs no background at all. Once they have a feel for what a matrix does, I define the determinant as the volume of the unit cube after the matrix acts on it, and we recover its defining properties from that picture: swapping two rows does not change the volume, just as swapping width and height does not. A whole chapter of results then follows from one idea, volume, and students can hold it in their minds as a single picture rather than a list of rules.

When I teach analysis, I pair the standard textbook with *Analysis by Its History*, so that each definition arrives with the problem it was invented to solve, and I use visual books such as *Visual Complex Analysis* wherever they exist.

MEET STUDENTS WHERE THEY ARE

I get to know my students and bend the material toward what they care about. When a section is full of biology or psychology students, I teach p-hacking: what goes wrong when you keep analysing data until something looks significant. I do research in biology and medicine as well as mathematics, so this is not a token application; it is the statistics that decides whether a drug works.

In remote discussion sections I split students into small groups to work problems together and moved between the rooms to check in, so that nobody sat stuck alone.

I also love working one-on-one. In Northwestern's Directed Reading Program (2020–2023) I met weekly with an undergraduate to read a topic of their choosing, ending with a talk to their peers. At the University of Chicago REU (2015–2017) I gave weekly lectures and guided each student through an individual research project. With Math Circles on the South Side of Chicago (2016–2017), I taught weekly mathematics to children aged six to twelve, and in Münster I designed hands-on workshops for children for Mathe trifft Kunst (Math Meets Art, 2025). The same approach works at every level: a set of puzzles to

play with, an object of direct interest, a question to guide the play, and a sequence of resources from my website, rin.io/resources.

MATHEMATICS IS A HUMAN ACTIVITY

Students learn better when they feel safe enough to be wrong out loud. My students noticed this. In my course evaluations for the honors linear algebra sequence (Spring 2021, 23 of 26 responding), I averaged 5.65 out of 6 for interest in teaching and 5.48 for clarity. One student wrote that I "made it such an inclusive, comfortable space"; another, that I give "feedback in an encouraging way."

Outside the classroom I look after the conditions that make learning possible. I founded Northwestern's Mental Health Math Buddy Program in 2021, which pairs first- and second-year graduate students with older students for quarterly check-ins, and I established a fund that lets graduate students take part in activities they could not otherwise afford. As a mentor in the Causeway Postbaccalaureate Program and the graduate school's diversity peer mentorship, I helped students from complex circumstances find their way into and through graduate school.

I also write for people who will never sit in my classroom. My blog, *Good Fibrations*, explains modern mathematics with drawings and in a chatty voice, because getting comfortable with these tools usually requires a professor who will spend hours with you, and most people never have one. A reader who used it to prepare for graduate school during the pandemic wrote that it "makes learning feel more safe and less hostile." My comic *re-search* is about the joy of understanding something for the first time, for readers with no mathematics at all.

I want anyone, whatever their background, to get both the methods of mathematics and the immersive feeling of doing it.